

Z_N Ring Delta-Pinning Mode Structure:

Verification of $\cos(N\theta)$ as Leading Anisotropic Mode

EDC Project

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Abstract

We analyze the eigenmode structure of a ring with N identical delta-function pinning sites at Z_N fixed points. Using group-theoretic decomposition and explicit Fourier analysis, we prove that the leading anisotropic eigenmode has angular dependence $\cos(N\theta)$. This verifies the mode ansatz used in the $a/c = 1/N$ derivation and the k -channel correction formula.

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1 Model Setup

1.1 Linearized Energy Functional

Definition 1.1 (Quadratic Anchor Potential). [\[Der\]](#) For linearization around equilibrium u_0 , we take:

$$W(u) = \frac{\kappa}{2}(u - u_0)^2 \quad (1)$$

where $\kappa = W''(u_0) > 0$ is the local stiffness.

Definition 1.2 (Linearized Energy Functional). [\[Der\]](#) The energy functional for small perturbations $v = u - u_0$:

$$E[v] = \frac{T}{2} \int_0^{2\pi} (v')^2 d\theta + \frac{\lambda\kappa}{2} \sum_{n=0}^{N-1} v(\theta_n)^2 \quad (2)$$

where $\theta_n = 2\pi n/N$ are the Z_N fixed points.

1.2 Euler-Lagrange Equation

Proposition 1.3 (Static Equation with Delta Sources). [\[Der\]](#) The Euler-Lagrange equation for $E[v]$ is:

$$-Tv''(\theta) + \lambda\kappa \sum_{n=0}^{N-1} v(\theta_n) \delta(\theta - \theta_n) = 0 \quad (3)$$

Proof. Variation of (2):

$$\delta E = T \int_0^{2\pi} v' \delta v' d\theta + \lambda\kappa \sum_{n=0}^{N-1} v(\theta_n) \delta v(\theta_n) \quad (4)$$

$$= -T \int_0^{2\pi} v'' \delta v d\theta + \lambda\kappa \sum_{n=0}^{N-1} v(\theta_n) \int_0^{2\pi} \delta(\theta - \theta_n) \delta v d\theta \quad (5)$$

Setting $\delta E = 0$ for all δv gives (3). □

1.3 Operator Form

Definition 1.4 (Linear Operator). [\[Der\]](#) Define the linear operator:

$$\mathcal{L} = -T \frac{d^2}{d\theta^2} + \lambda\kappa \sum_{n=0}^{N-1} \delta(\theta - \theta_n) P_n \quad (6)$$

where P_n is the projection (evaluation) at θ_n : $(P_n f) = f(\theta_n)$.

The eigenvalue problem is:

$$\mathcal{L}v = \mu v \quad (7)$$

2 Group-Theoretic Decomposition

2.1 Z_N Symmetry of the Operator

Lemma 2.1 (Z_N Invariance). *[Der] The operator $\mathcal{L}$ commutes with the Z_N rotation $R_{2\pi/N}$:*

$$[\mathcal{L}, R_{2\pi/N}] = 0 \quad (8)$$

Proof. The Laplacian $-d^2/d\theta^2$ is rotationally invariant. The pinning term $\sum_n \delta(\theta - \theta_n)$ is invariant under $\theta \rightarrow \theta + 2\pi/N$ because the set $\{\theta_n\}$ maps to itself. $\square$

Corollary 2.2 (Mode Classification). *[Der] Eigenfunctions of $\mathcal{L}$ can be classified by Z_N irreducible representations. For Z_N , the irreps are labeled by $\ell = 0, 1, \dots, N-1$:*

$$R_{2\pi/N} \psi_\ell = e^{2\pi i \ell / N} \psi_\ell \quad (9)$$

2.2 Fourier Basis and Z_N Sectors

Proposition 2.3 (Fourier Decomposition into Z_N Sectors). *[Der] The Fourier mode $e^{im\theta}$ belongs to Z_N sector $\ell = m \bmod N$:*

$$R_{2\pi/N} e^{im\theta} = e^{im(\theta+2\pi/N)} = e^{2\pi i m / N} e^{im\theta} \quad (10)$$

Therefore, modes in sector ℓ have the form:

$$e^{i(\ell+kN)\theta}, \quad k \in \mathbb{Z} \quad (11)$$

Key Insight: Sector Structure

The Z_N symmetry decomposes the Hilbert space into N sectors:

$$\text{Sector } \ell = 0 : \quad \{1, e^{\pm iN\theta}, e^{\pm 2iN\theta}, \dots\} = \{1, \cos(N\theta), \cos(2N\theta), \dots\} \quad (12)$$

$$\text{Sector } \ell = 1 : \quad \{e^{\pm i\theta}, e^{\pm i(N+1)\theta}, \dots\} \quad (13)$$

$$\vdots \quad (14)$$

$$\text{Sector } \ell = N-1 : \quad \{e^{\pm i(N-1)\theta}, e^{\pm i(2N-1)\theta}, \dots\} \quad (15)$$

The $\ell = 0$ sector contains all Z_N -symmetric functions.

3 Pinning Term Coupling Analysis

3.1 Pinning Strength for Different Modes

Theorem 3.1 (Mode Coupling to Pinning Sites). *[Der] For a Fourier mode $e^{im\theta}$, the coupling to the pinning term is:*

$$\sum_{n=0}^{N-1} e^{im\theta_n} = \sum_{n=0}^{N-1} e^{2\pi i m n / N} = \begin{cases} N & \text{if } m \equiv 0 \pmod{N} \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

Proof. This is the standard geometric series:

$$\sum_{n=0}^{N-1} \omega^{mn} = \frac{1 - \omega^{mN}}{1 - \omega^m} \quad \text{where } \omega = e^{2\pi i/N} \quad (17)$$

If $m \equiv 0 \pmod{N}$, then $\omega^m = 1$ and all terms equal 1. Otherwise, $\omega^{mN} = 1$ but $\omega^m \neq 1$, so the sum vanishes. $\square$

Corollary 3.2 (Selective Coupling). *[Der] The pinning term couples only to Z_N -symmetric modes (sector $\ell = 0$).*

For real modes:

- $\cos(mN\theta)$: maximally coupled (all sites contribute coherently)
- $\cos(m\theta)$ with $m \not\equiv 0 \pmod{N}$: zero net coupling

4 Selection Lemma: $\cos(N\theta)$ as Leading Mode

4.1 Unpinned Spectrum

Proposition 4.1 (Free Ring Spectrum). *[Der] Without pinning ($\lambda = 0$), the eigenvalues of $-Td^2/d\theta^2$ are:*

$$\mu_m^{(0)} = Tm^2, \quad m = 0, 1, 2, \dots \quad (18)$$

with eigenfunctions $\cos(m\theta)$, $\sin(m\theta)$ (degenerate for $m > 0$).

4.2 Effect of Pinning: Perturbative Analysis

Theorem 4.2 (Mode Lifting by Pinning). *[Der] Consider the dimensionless pinning strength:*

$$\rho = \frac{\lambda\kappa}{T} \quad (19)$$

*(a) Non- Z_N -symmetric modes ($m \not\equiv 0 \pmod{N}$): These are **unaffected** by pinning (zero net coupling). Their eigenvalues remain $\mu_m = Tm^2$.*

*(b) Z_N -symmetric modes ($m = kN$): These are **lifted** by the pinning term.*

Proof. From Corollary 3.2, modes with $m \not\equiv 0 \pmod{N}$ have zero matrix element with the pinning perturbation.

For Z_N -symmetric modes, the pinning term acts as a positive definite perturbation, raising their eigenvalues. $\square$

4.3 The Selection Lemma

Lemma 4.3 (Selection Lemma). *[Der] Among all anisotropic modes (excluding the constant $m = 0$), the mode with the **lowest gradient energy** that couples to the pinning sites is:*

$$\boxed{v(\theta) = \cos(N\theta)} \quad (20)$$

*This is the **first** Z_N -symmetric harmonic above the constant.*

Proof. Z_N -symmetric harmonics have $m = kN$ for $k = 0, 1, 2, \dots$

- $k = 0$: constant mode (isotropic, $m = 0$)
- $k = 1$: $\cos(N\theta)$ with gradient energy $\propto N^2$
- $k = 2$: $\cos(2N\theta)$ with gradient energy $\propto 4N^2$
- etc.

The lowest-energy anisotropic mode that couples to the anchors is $\cos(N\theta)$. □

4.4 Physical Interpretation

Why $\cos(N\theta)$ Dominates

Two competing effects:

1. **Gradient energy** favors low harmonics: $E_{\text{grad}} \propto m^2$
2. **Pinning coupling** requires $m \equiv 0 \pmod{N}$

Resolution:

- Modes with $m < N$ (e.g., $\cos \theta$, $\cos 2\theta$) have lower gradient energy but **zero coupling** to the pinning sites.
- The first mode that **both** couples to pinning **and** is anisotropic is $m = N$, i.e., $\cos(N\theta)$.

Conclusion: $\cos(N\theta)$ is the unique leading anisotropic mode.

5 Explicit Eigenvalue Calculation

5.1 Ansatz in Z_N -Symmetric Sector

Proposition 5.1 (Z_N -Symmetric Mode Eigenvalue). [Der] For the mode $v(\theta) = \cos(N\theta)$:

Gradient contribution:

$$-Tv'' = TN^2 \cos(N\theta) \tag{21}$$

Pinning contribution: At $\theta_n = 2\pi n/N$: $\cos(N\theta_n) = \cos(2\pi n) = 1$.

The equation $\mathcal{L}v = \mu v$ in distributional form:

$$TN^2 \cos(N\theta) + \lambda\kappa \sum_{n=0}^{N-1} \delta(\theta - \theta_n) = \mu \cos(N\theta) \tag{22}$$

Taking the inner product with $\cos(N\theta)$:

$$TN^2 \cdot \pi + \lambda\kappa \cdot N = \mu \cdot \pi \tag{23}$$

Therefore:

$$\boxed{\mu_N = TN^2 + \frac{\lambda\kappa N}{\pi} = TN^2 \left(1 + \frac{\rho}{\pi N}\right)} \tag{24}$$

5.2 Comparison with Lower Modes

Proposition 5.2 (Non-Coupled Mode Eigenvalues). [\[Der\]](#) For $m \not\equiv 0 \pmod{N}$, the eigenvalue is unchanged:

$$\mu_m = Tm^2 \quad (m = 1, 2, \dots, N-1) \quad (25)$$

Eigenvalue Ordering

Mode	Eigenvalue	$\mathbf{Z}_N$ Sector	Pinning
$m = 0$ (const)	0	$\ell = 0$	coupled
$m = 1$	T	$\ell = 1$	not coupled
$m = 2$	$4T$	$\ell = 2$	not coupled
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$m = N-1$	$T(N-1)^2$	$\ell = N-1$	not coupled
$m = N$	$TN^2(1 + \rho/\pi N)$	$\ell = 0$	coupled
$m = N+1$	$T(N+1)^2$	$\ell = 1$	not coupled

6 Regime Analysis

6.1 Weak Pinning Regime

Definition 6.1 (Weak Pinning). [\[Der\]](#)

$$\rho = \frac{\lambda\kappa}{T} \ll N^2 \quad (26)$$

Proposition 6.2 (Weak Pinning Result). [\[Der\]](#) In the weak pinning regime:

- The mode structure is dominated by gradient energy.
- $\cos(N\theta)$ is the **only** $\mathbf{Z}_N$ -symmetric anisotropic mode that couples to anchors at leading order.
- Lower harmonics ($m < N$) do not couple, so cannot be excited by anchor forcing.

6.2 Strong Pinning Regime

Definition 6.3 (Strong Pinning). [\[Der\]](#)

$$\rho = \frac{\lambda\kappa}{T} \gg N^2 \quad (27)$$

Proposition 6.4 (Strong Pinning Result). [\[Der\]](#) In the strong pinning regime:

- The field is pinned near $v = 0$ at all anchor sites.
- Between anchors, the field interpolates smoothly.
- The dominant mode structure **still** has $\mathbf{Z}_N$ periodicity.
- $\cos(N\theta)$ remains the leading anisotropic component.

7 Conclusion

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Result: Under the Z_N delta-pinning model, $\cos(N\theta)$ is the **unique leading anisotropic mode**.

Proof summary:

1. Pinning couples **only** to Z_N -symmetric modes (Theorem 3.1)
2. The lowest Z_N -symmetric anisotropic harmonic is $m = N$ (Lemma 4.3)
3. Therefore, $\cos(N\theta)$ is the first mode that is both anisotropic and coupled to the anchors.

Conditions:

- Identical anchors at Z_N fixed points (verified in Israel derivation)
- Quadratic $W(u)$ near equilibrium (standard linearization)
- Ring geometry with periodic boundary conditions

This validates the ansatz $u(\theta) = u_0 + a_1 \cos(N\theta)$ used in the $a/c = 1/N$ derivation.

8 Epistemic Status

Result	Status	Comment
Z_N sector decomposition	[Der]	Group theory
Pinning couples only to $\ell = 0$ sector	[Der]	Geometric series identity
$\cos(N\theta)$ is lowest anisotropic in $\ell = 0$	[Der]	By construction
Selection Lemma	[Der]	Follows from above
Eigenvalue formula (24)	[Der]	Direct calculation
Mapping to 5D brane physics	[Dc]	Requires bulk solution
$W(u)$ being quadratic near equilibrium	[Dc]	Assumed, not derived

Summary: Within the delta-pinning ring model, all results are [Der]. The mapping to full 5D physics remains [Dc].